

AI-Powered Academic Assistant using Retrieval-Augmented Generation (RAG)

Problem Statement

Students spend significant time searching through notes, PDFs, textbooks, question papers, and study materials to find relevant academic information. Traditional search methods are often time-consuming and may not provide accurate, context-aware responses. To address this challenge, the proposed system aims to develop an AI-Powered Academic Assistant using Retrieval-Augmented Generation (RAG). The system will retrieve relevant information from academic documents and generate accurate, context-aware answers to student queries. By combining document retrieval with large language models, the assistant can improve learning efficiency, reduce information search time, and provide personalized academic support.

Objectives

- Retrieve relevant information from academic documents.
- Provide accurate and context-aware answers to student questions.
- Reduce the time spent searching study materials.
- Improve learning and knowledge accessibility through AI.